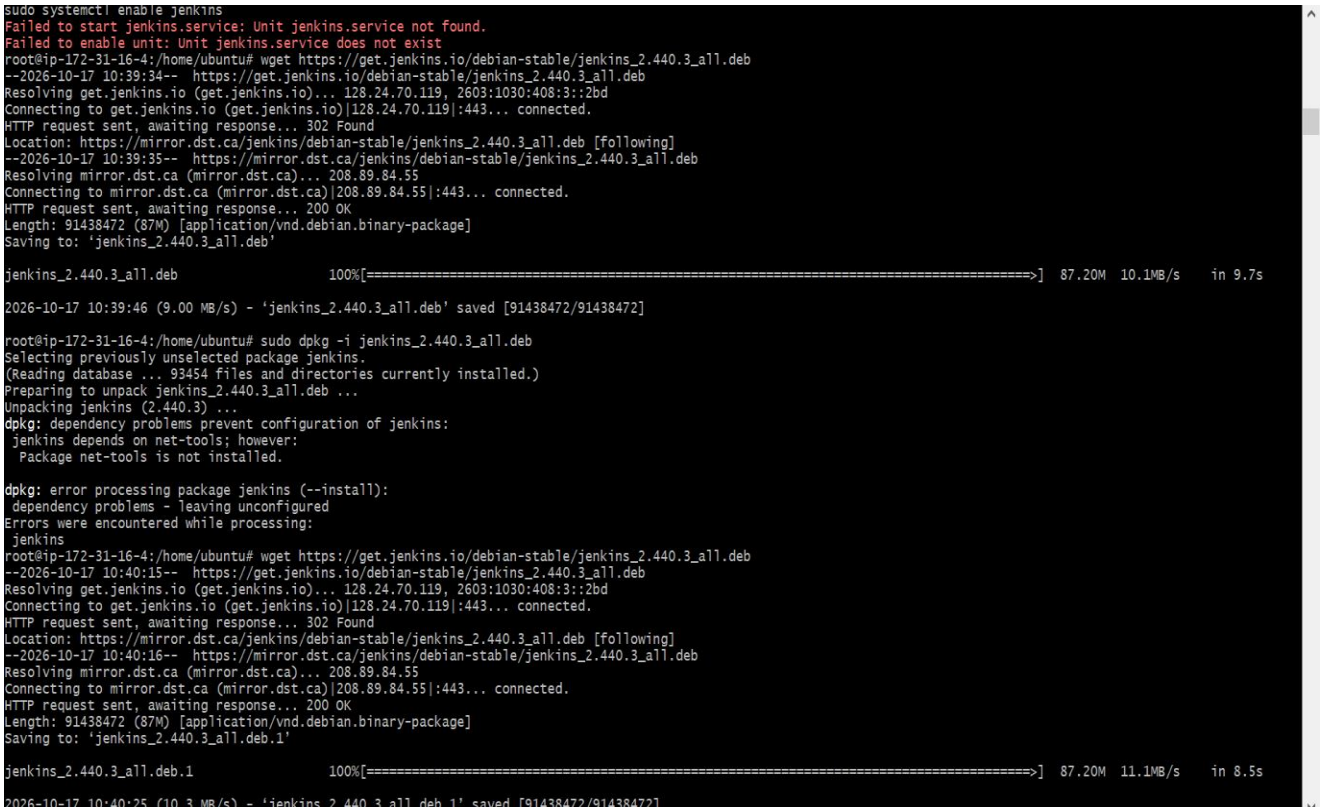
INSTALLATIONS OF JENKINS TOMCAT AND DOCKER

Notes: ***Create a Jira ticket for this assignment, Provide detailed step-by-step instructions for the installation process along with proper evidence (Outputs, and screenshots). You may refer to YouTube tutorials or Google resources for guidance, and submit the challenge completion document by EOD.***

1. Launch one aws ec2 and install jenkins.
2. Setup Apache Tomcat on Ubuntu Virtual Machine
3. Setup Docker Desktop on your local windows.

Log off only after completing all pending tasks and challenges till date.

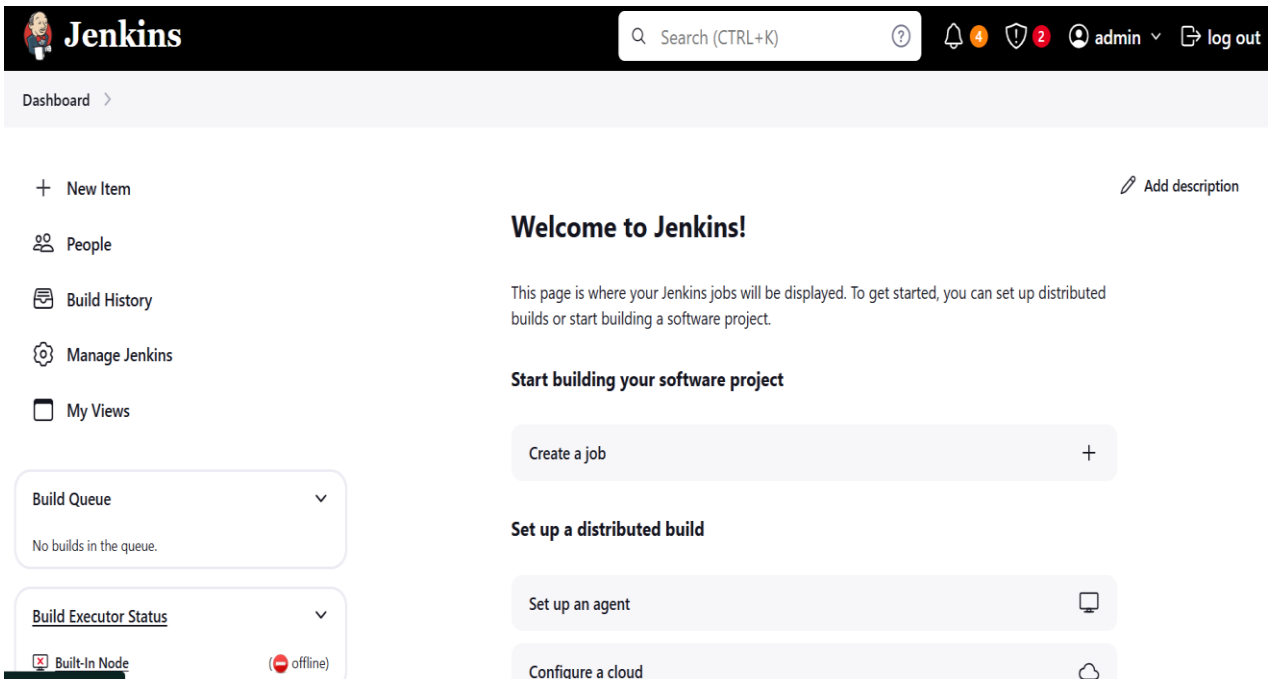
1. Launch one aws ec2 and install jenkins.

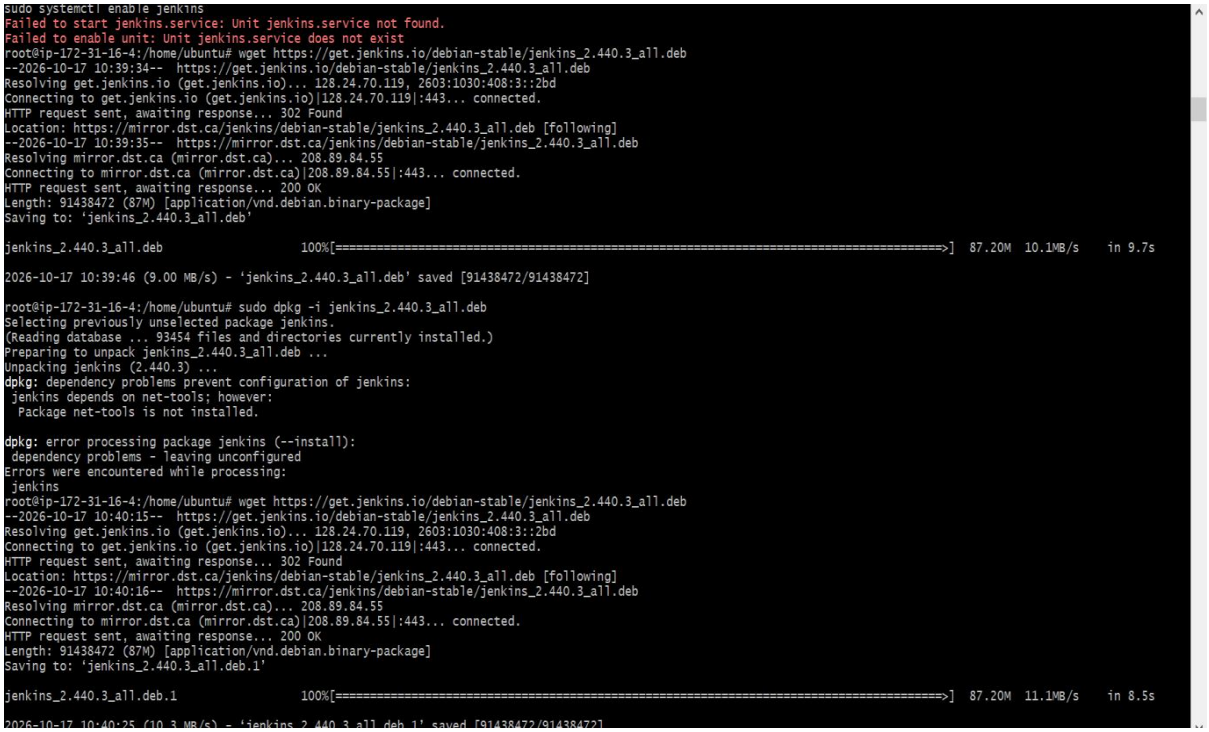


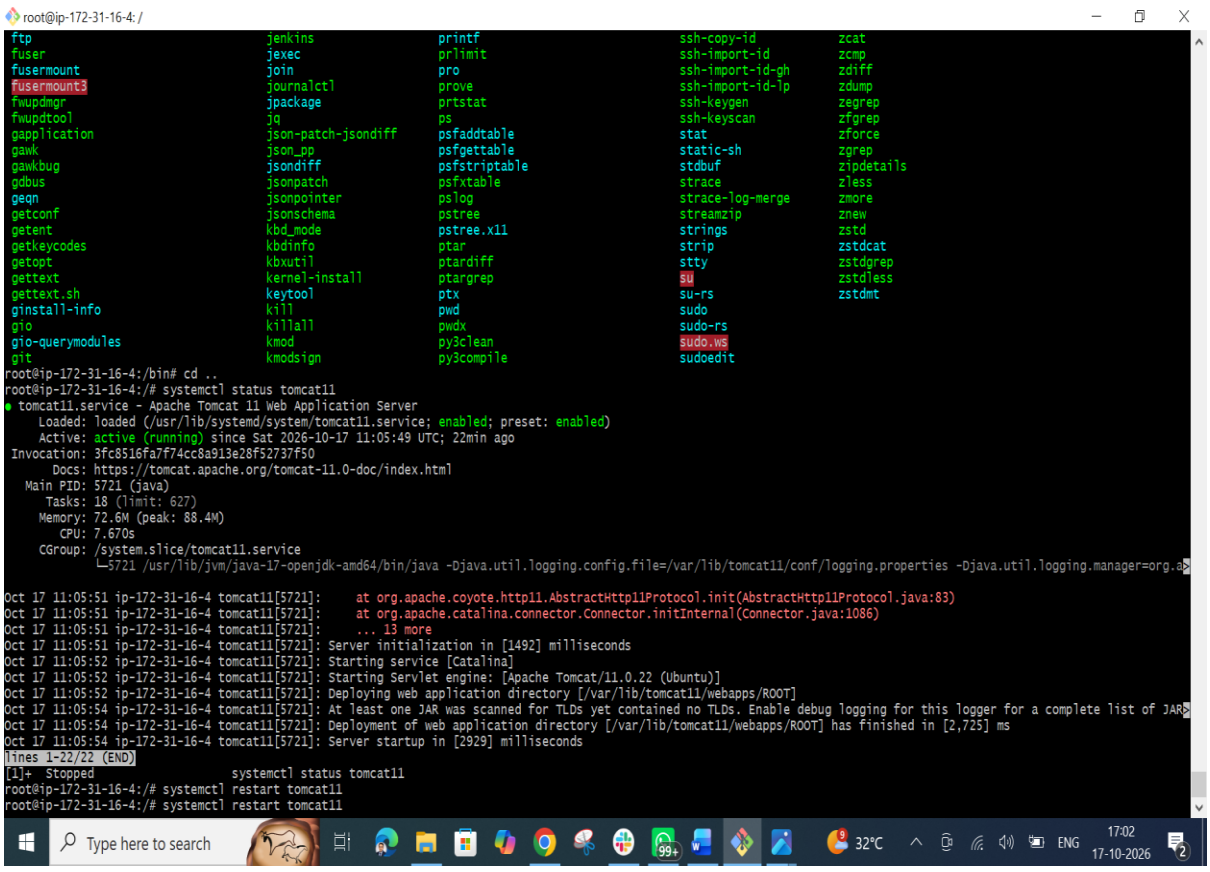
```
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
root@ip-172-31-16-4:/home/ubuntu# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-17 10:42:24 UTC; 1min 33s ago
 Invocation: b85d19808dbf4a5cadf1fb5863119d07
   Main PID: 3402 (java)
      Tasks: 39 (limit: 627)
     Memory: 305.4M (peak: 327.5M)
        CPU: 1min 40.669s
     CGroup: /system.slice/jenkins.service
             └─3402 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

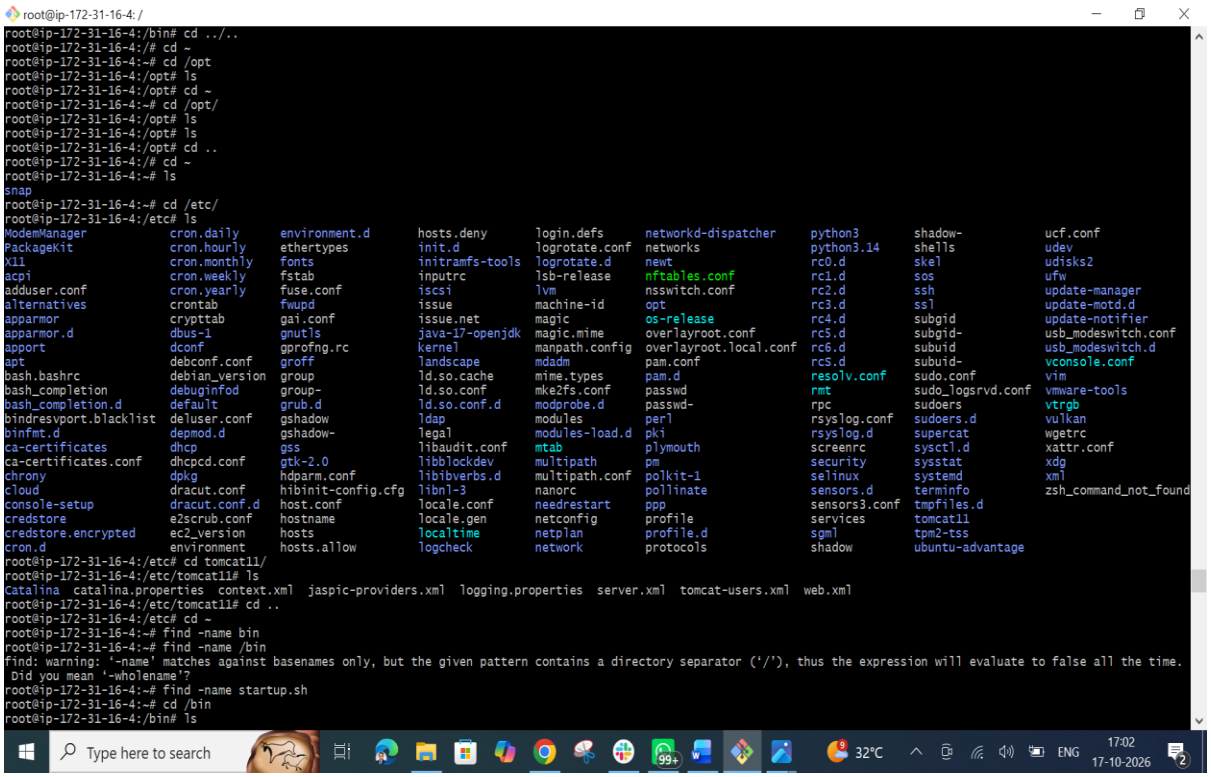
Oct 17 10:41:18 ip-172-31-16-4 jenkins[3402]: This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Oct 17 10:41:18 ip-172-31-16-4 jenkins[3402]: ************************************************************
Oct 17 10:41:18 ip-172-31-16-4 jenkins[3402]: ************************************************************
Oct 17 10:41:18 ip-172-31-16-4 jenkins[3402]: ************************************************************
Oct 17 10:42:24 ip-172-31-16-4 jenkins[3402]: 2026-10-17 10:42:24.637+0000 [id=33]        INFO        jenkins.InitReactorRunner$1#onAttained: Completed initialization
Oct 17 10:42:24 ip-172-31-16-4 jenkins[3402]: 2026-10-17 10:42:24.673+0000 [id=24]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Oct 17 10:42:24 ip-172-31-16-4 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 17 10:42:26 ip-172-31-16-4 jenkins[3402]: 2026-10-17 10:42:26.265+0000 [id=49]        INFO        h.m.DownloadService$Downloadable#load: Obtained the updated data file>
Oct 17 10:42:26 ip-172-31-16-4 jenkins[3402]: 2026-10-17 10:42:26.266+0000 [id=49]        INFO        hudson.util.Retrier#start: Performed the action check updates server >
Oct 17 10:42:29 ip-172-31-16-4 jenkins[3402]: 2026-10-17 10:42:29.772+0000 [id=68]        WARNING         h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOrOnline: Making Bui>
root@ip-172-31-16-4:/home/ubuntu# cd /var/lib/jenkins/secrets/initialAdminPassword
-bash: cd: /var/lib/jenkins/secrets/initialAdminPassword: Not a directory
root@ip-172-31-16-4:/home/ubuntu# cd ..
root@ip-172-31-16-4:/home# cd /var/lib/jenkins/secrets/initialAdminPassword
-bash: cd: /var/lib/jenkins/secrets/initialAdminPassword: Not a directory
root@ip-172-31-16-4:/home# cd ..
root@ip-172-31-16-4:/# cd /var/lib/jenkins/secrets/initialAdminPassword
-bash: cd: /var/lib/jenkins/secrets/initialAdminPassword: Not a directory
root@ip-172-31-16-4:/# cd ..
root@ip-172-31-16-4:/# ls
bin  boot  dev  etc  home  lib  lib64  lost+found  media  mnt  opt  proc  root  run  sbin  snap  srv  sys  tmp  usr  var
root@ip-172-31-16-4:/# cd var
root@ip-172-31-16-4:/var# cd lib
root@ip-172-31-16-4:/var/lib# cd jenkins
root@ip-172-31-16-4:/var/lib/jenkins# cd secret
-bash: cd: secret: No such file or directory
root@ip-172-31-16-4:/var/lib/jenkins# cd secrets
root@ip-172-31-16-4:/var/lib/jenkins/secrets# cd initialAdminPassword
-bash: cd: initialAdminPassword: Not a directory
root@ip-172-31-16-4:/var/lib/jenkins/secrets# cat initialAdminPassword
0efe77267cb74dad8602602890ce14e5
root@ip-172-31-16-4:/var/lib/jenkins/secrets# client_loop: send disconnect: Connection reset by peer

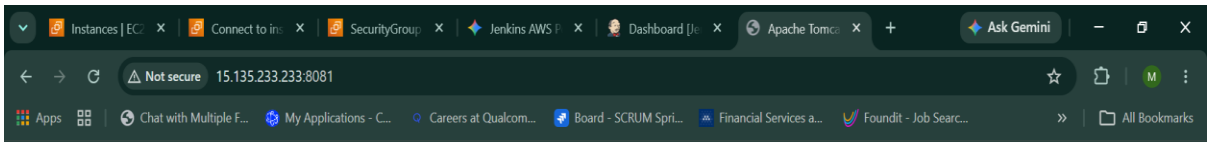
mdtan@DESKTOP-T7V22P3 MINGW64 ~/Downloads
$
```




sudo systemctl enable jenkins
Failed to start jenkins.service: Unit jenkins.service not found.
Failed to enable unit: Unit jenkins.service does not exist
root@ip-172-31-16-4:/home/ubuntu# wget https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb
--2026-10-17 10:39:34-- https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb
Resolving get.jenkins.io (get.jenkins.io)... 128.24.70.119, 2603:1030:408:3::2bd
Connecting to get.jenkins.io (get.jenkins.io)|128.24.70.119|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://mirror.dst.ca/jenkins/debian-stable/jenkins_2.440.3_all.deb [following]
--2026-10-17 10:39:35-- https://mirror.dst.ca/jenkins/debian-stable/jenkins_2.440.3_all.deb
Resolving mirror.dst.ca (mirror.dst.ca)... 208.89.84.55
Connecting to mirror.dst.ca (mirror.dst.ca)|208.89.84.55|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 91438472 (87M) [application/vnd.debian.binary-package]
Saving to: 'jenkins_2.440.3_all.deb'
jenkins_2.440.3_all.deb 100%[=>] 87.20M 10.1MB/s in 9.7s
2026-10-17 10:39:46 (9.00 MB/s) - 'jenkins_2.440.3_all.deb' saved [91438472/91438472]
root@ip-172-31-16-4:/home/ubuntu# sudo dpkg -i jenkins_2.440.3_all.deb
Selecting previously unselected package jenkins.
(Reading database ... 93454 files and directories currently installed.)
Preparing to unpack jenkins_2.440.3_all.deb ...
Unpacking jenkins (2.440.3) ...
dpkg: dependency problems prevent configuration of jenkins:
 jenkins depends on net-tools; however:
  Package net-tools is not installed.
dpkg: error processing package jenkins (--install):
 dependency problems - leaving unconfigured
Errors were encountered while processing:
 jenkins
root@ip-172-31-16-4:/home/ubuntu# wget https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb
--2026-10-17 10:40:15-- https://get.jenkins.io/debian-stable/jenkins_2.440.3_all.deb
Resolving get.jenkins.io (get.jenkins.io)... 128.24.70.119, 2603:1030:408:3::2bd
Connecting to get.jenkins.io (get.jenkins.io)|128.24.70.119|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://mirror.dst.ca/jenkins/debian-stable/jenkins_2.440.3_all.deb [following]
--2026-10-17 10:40:16-- https://mirror.dst.ca/jenkins/debian-stable/jenkins_2.440.3_all.deb
Resolving mirror.dst.ca (mirror.dst.ca)... 208.89.84.55
Connecting to mirror.dst.ca (mirror.dst.ca)|208.89.84.55|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 91438472 (87M) [application/vnd.debian.binary-package]
Saving to: 'jenkins_2.440.3_all.deb.1'
jenkins_2.440.3_all.deb.1 100%[=>] 87.20M 11.1MB/s in 8.5s


root@ip-172-31-16-4: /
ftp jenkins printf ssh-copy-id zcat
fuser jexec prlimit ssh-import-id zcmp
fusermount join pro ssh-import-id-gh zdiff
fusermount3 journalctl prove ssh-import-id-lp zdump
fwupdmgr jpackage prtstat ssh-keygen zegrep
fwupdtool jq ps ssh-keyscan zfgrep
gapplication json-patch-jsondiff psfaddtable stat zforce
gawk json_pp psfgettable static-sh zgrep
gawkbug jsondiff psfstriptable stdbuf zipdetails
gdbus jsonpatch psfxtable strace zless
geqn jsonpointer pslog strace-log-merge zmore
getconf jsonschema pstree streamzip znew
getent kbd_mode pstree.x11 strings zstd
getkeycodes kbdinfo ptar strip zstdcat
getopt kbxutil ptardiff stty zstdgrep
gettext kernel-install ptargrep su zstdless
gettext.sh keytool ptx su-rs zstdmt
ginstall-info kill pwd sudo
gio killall pwdx sudo-rs
gio-querymodules kmod py3clean sudo.ws
git kmodsign py3compile sudoedit
root@ip-172-31-16-4:/bin# cd ..
root@ip-172-31-16-4:/# systemctl status tomcat11
● tomcat11.service - Apache Tomcat 11 Web Application Server
Loaded: loaded (/usr/lib/systemd/system/tomcat11.service; enabled; preset: enabled)
Active: active (running) since Sat 2026-10-17 11:05:49 UTC; 22min ago
Invocation: 3fc8516fa7f74cc8a913e28f52737f50
Docs: https://tomcat.apache.org/tomcat-11.0-doc/index.html
Main PID: 5721 (java)
Tasks: 18 (limit: 627)
Memory: 72.6M (peak: 88.4M)
CPU: 7.670s
CGroup: /system.slice/tomcat11.service
└─5721 /usr/lib/jvm/java-17-openjdk-amd64/bin/java -Djava.util.logging.config.file=/var/lib/tomcat11/conf/logging.properties -Djava.util.logging.manager=org.a
Oct 17 11:05:51 ip-172-31-16-4 tomcat11[5721]: at org.apache.coyote.http11.AbstractHttp11Protocol.init(AbstractHttp11Protocol.java:83)
Oct 17 11:05:51 ip-172-31-16-4 tomcat11[5721]: at org.apache.catalina.connector.Connector.initInternal(Connector.java:1086)
Oct 17 11:05:51 ip-172-31-16-4 tomcat11[5721]: ... 13 more
Oct 17 11:05:51 ip-172-31-16-4 tomcat11[5721]: Server initialization in [1492] milliseconds
Oct 17 11:05:52 ip-172-31-16-4 tomcat11[5721]: Starting service [Catalina]
Oct 17 11:05:52 ip-172-31-16-4 tomcat11[5721]: Starting Servlet engine: [Apache Tomcat/11.0.22 (Ubuntu)]
Oct 17 11:05:52 ip-172-31-16-4 tomcat11[5721]: Deploying web application directory [/var/lib/tomcat11/webapps/ROOT]
Oct 17 11:05:54 ip-172-31-16-4 tomcat11[5721]: At least one JAR was scanned for TLDs yet contained no TLDs. Enable debug logging for this logger for a complete list of JAR
Oct 17 11:05:54 ip-172-31-16-4 tomcat11[5721]: Deployment of web application directory [/var/lib/tomcat11/webapps/ROOT] has finished in [2,725] ms
Oct 17 11:05:54 ip-172-31-16-4 tomcat11[5721]: Server startup in [2929] milliseconds
lines 1-22/22 (END)
[1]+ Stopped systemctl status tomcat11
root@ip-172-31-16-4:/# systemctl restart tomcat11
root@ip-172-31-16-4:/# systemctl restart tomcat11
Type here to search
32°C
ENG
17:02
17-10-2026





It works !

If you're seeing this page via a web browser, it means you've setup Tomcat successfully. Congratulations!

This is the default Tomcat home page. It can be found on the local filesystem at: `/var/lib/tomcat11/webapps/ROOT/index.html`

Tomcat veterans might be pleased to learn that this system instance of Tomcat is installed with `CATALINA_HOME` in `/usr/share/tomcat11` and `CATALINA_BASE` in `/var/lib/tomcat11`, following the rules from `/usr/share/doc/tomcat11-common/RUNNING.txt.gz`.

You might consider installing the following packages, if you haven't already done so:

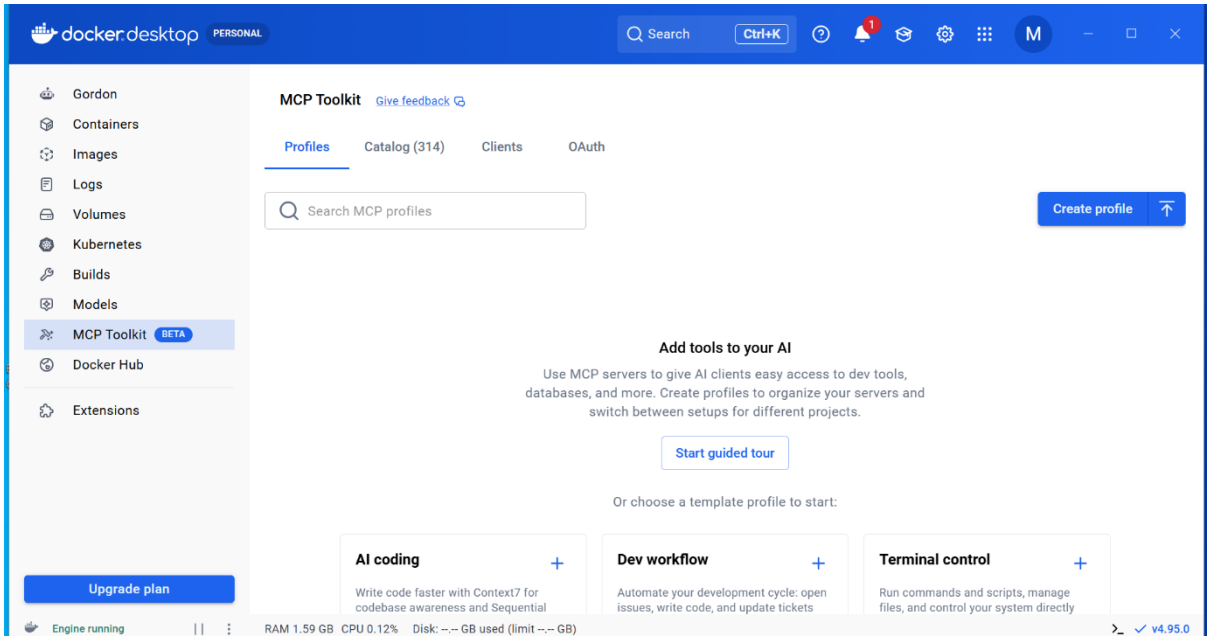
tomcat11-docs: This package installs a web application that allows to browse the Tomcat 11 documentation locally. Once installed, you can access it by clicking here.

tomcat11-examples: This package installs a web application that allows to access the Tomcat 11 Servlet and JSP examples. Once installed, you can access it by clicking here.

tomcat11-admin: This package installs two web applications that can help managing this Tomcat instance. Once installed, you can access the manager webapp and the host-manager webapp.

NOTE: For security reasons, using the manager webapp is restricted to users with role "manager-gui". The host-manager webapp is restricted to users with role "admin-gui". Users are defined in `/etc/tomcat11/tomcat-users.xml`.




docker desktop
PERSONAL
Search
Ctrl+K
M
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
MCP Toolkit
BETA
Docker Hub
Extensions
Upgrade plan
Engine running
MCP Toolkit
Give feedback
Profiles
Catalog (314)
Clients
OAuth
Search MCP profiles
Create profile
Add tools to your AI
Use MCP servers to give AI clients easy access to dev tools, databases, and more. Create profiles to organize your servers and switch between setups for different projects.
Start guided tour
Or choose a template profile to start:
AI coding
Write code faster with Context7 for codebase awareness and Sequential
Dev workflow
Automate your development cycle: open issues, write code, and update tickets
Terminal control
Run commands and scripts, manage files, and control your system directly
RAM 1.59 GB CPU 0.12% Disk: --.-- GB used (limit --.-- GB)
v4.95.0